OUTPUT FILE

--- 1. LASSO Regression ---

Model: LASSO Regression

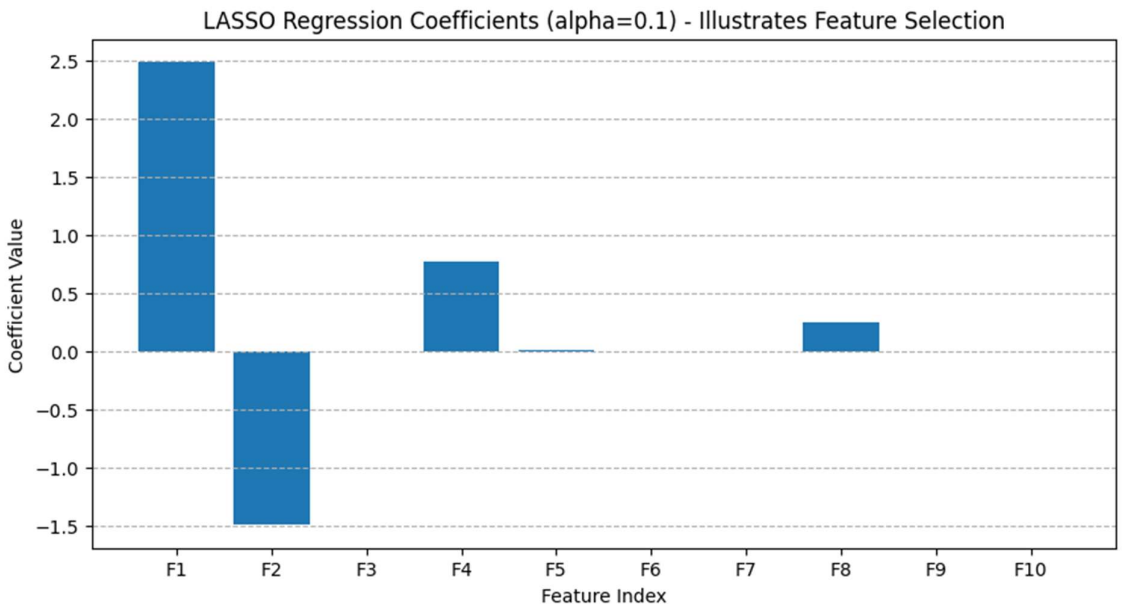
Alpha Hyperparameter: 0.1

Parameter Values (Coefficients):

[2.48496761 -1.49206889 -0. 0.77077978 0.01063494 0.

 -0. 0.24363542 -0. 0.]

Mean Squared Error (MSE) on Test Set: 0.2446



--- 2. K-means Clustering ---

Model: K-means Clustering

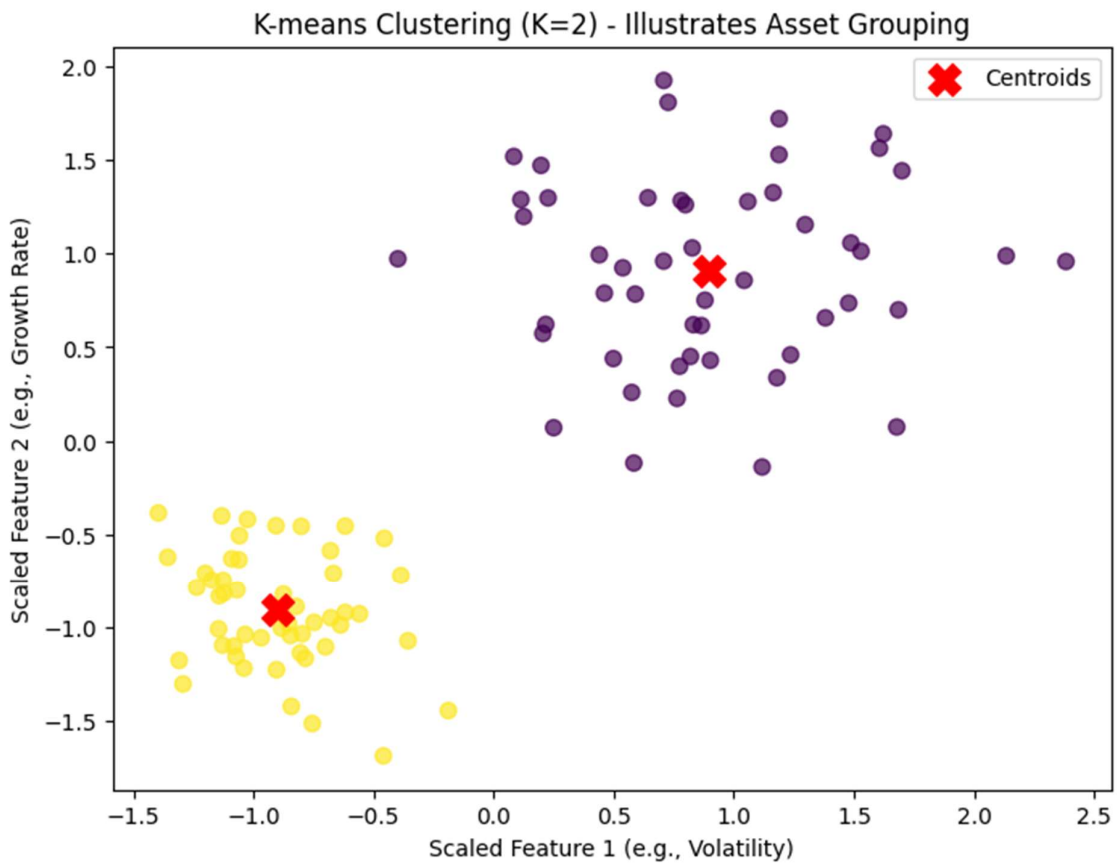
K Hyperparameter: 2

Parameter Values (Cluster Centers - Scaled):

[[0.89758627 0.91008317]

 [-0.89758627 -0.91008317]]

Inertia (Within-Cluster Sum of Squares): 36.61



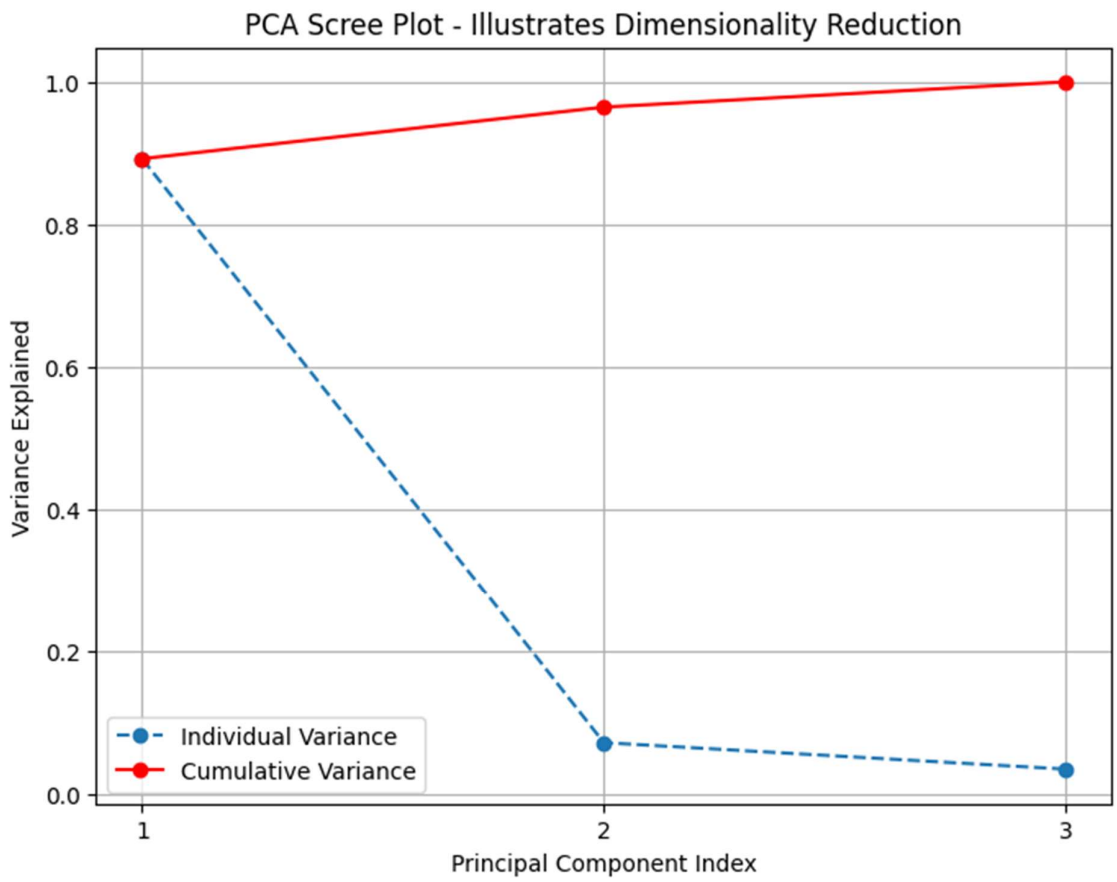
--- 3. Principal Components Analysis (PCA) ---

Model: PCA

n_components Hyperparameter: 3

Explained Variance Ratio by Component: [0.8921706 0.07240716 0.03542224]

Cumulative Explained Variance: [0.8921706 0.96457776 1.   ]

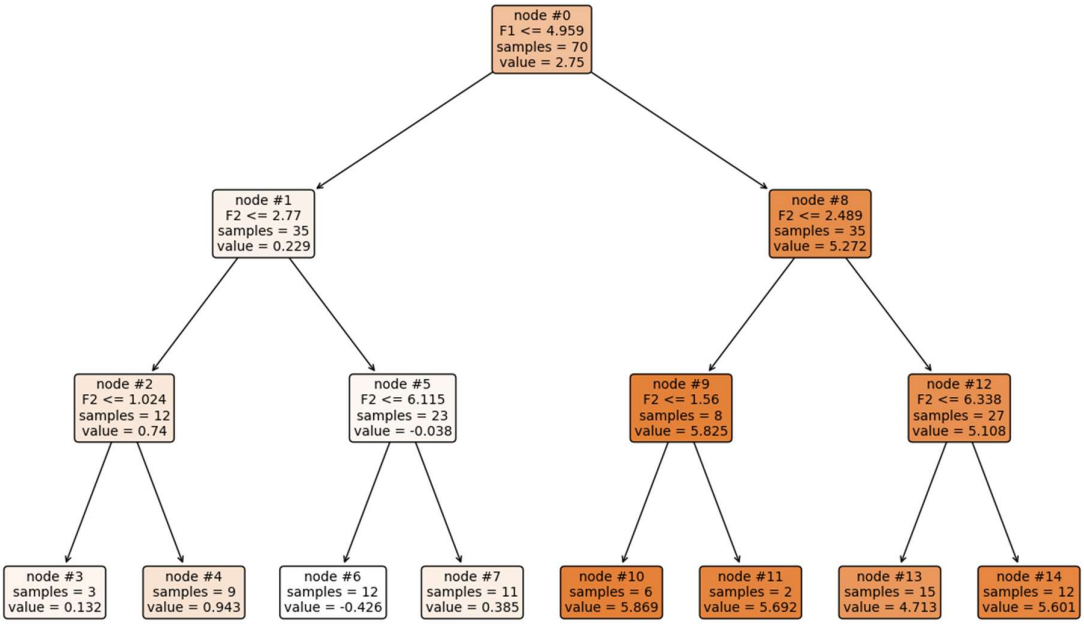


--- 4. Regression Trees ---

Model: Decision Tree Regressor

Max_Depth Hyperparameter: 3

Mean Squared Error (MSE) on Test Set: 0.2234

Feature Importances: [0.96004147 0.03995853 0.]

Regression Tree Flowchart (Max Depth=3) - Illustrates Decision Rules